6507765026
tessone@usc.edu

DEAN TESSONE

Summary

Ph.D. student in Molecular and Computational Biology developing deep-learning models for liquid biopsy. Research focuses on representation learning and weakly supervised learning, with applications in single-cell microscopy, genomics, disease monitoring, and biomarker discovery.

Education

University of Southern California – *Los Angeles, CA*

Ph.D. Molecular and Computational Biology **Fall 2023 – Present**

GPA: 4.0

Honors: Rockwell Dennis Hunt Award; 1st Place Poster Award, SBI2 2025; 1st Place Poster Award, USC Industry Day

B.S. Biological Sciences (Molecular, Cellular, & Developmental Biology) **May 2023**

GPA: 3.95

Awards: The Order of the Laurel and Palm (Top 24 in the Class of 2023), SCynergy award (Top Graduate, Biology Department), Renaissance Scholar, Phi Beta Kappa

Professional Experience

Tempus AI – *Chicago, IL*

Foundation Models Intern **June-August 2026**

- Designed and implemented Tempus' Genomics Foundation Model for patient-derived DNA, RNA, and ctDNA in the largest available multimodal dataset and model to date.

Convergent Science Institute in Cancer (Dr. Peter Kuhn Lab) – *Los Angeles, CA*

Ph.D. Student **Fall 2023-Present**

- Develop interpretable foundation models for cancer detection from circulating single-cell imaging data, with multimodal (CTCs and cfDNA) integration.
- Designed a novel rare-event detection algorithm that outperforms existing methods for detection of circulating tumor and tumor-associated cells in liquid biopsies.

Lead Undergraduate Researcher **2019-2023**

- Identified and characterized circulating cells and extracellular vesicles in peripheral blood from multiple myeloma patients for disease stratification and longitudinal monitoring.

Genentech – *South San Francisco, CA*

Oncology Biomarker Development Intern **May-August 2021**

- Applied unsupervised machine-learning methods to gene expression data from diffuse large B-cell lymphoma clinical trials to identify latent molecular programs/

Selected Publications

1. Murgoitio-Esandi, J., **Tessone, D**. et al. Unsupervised detection of rare events in liquid biopsy assays. npj Precis. Onc. **9**, 225 (2025).
2. Naghdloo, A.*, **Tessone, D.***, Nagaraju, R.M.* et al. Representation learning enables robust single cell phenotyping in whole slide liquid biopsy imaging. Sci Rep **15**, 36589 (2025). (Co-First Author)
3. **Tessone D,** et al. OpenIMC: an open-source platform for analyzing single-cell and spatial proteomics by Imaging Mass Cytomtetry. **Submitted to BMC Bioinformatics. 2026.**

Additional Skills

Python (PyTorch, ML/AI, Data Analysis + Visualization), R, Git, Distributed Training, Foundation Models, Unsupervised Learning, Computer Vision, Image Analysis, Single-cell analysis, cell culture, IF staining/assay development